

PROJECT CHARTER	
Project Name	AI Learning Environment (working name)
Date Produced	6 October 2026
Project Goals	Build a desktop study workspace that teaches from the course files already on a student's device, and test whether it can grow into a self-funded product. The project will: • Remove the repeated setup step: students get AI help on the file they have open, without uploading files or re-explaining the course every session. • Make answers verifiable: every explanation points to the page it came from. • Turn reading into practice: flashcards and quizzes are generated from the same passage, with no manual card-making. • Prove a sustainable model: a free tier on free AI models and a Pro subscription that pays for paid models, so the product can run without outside funding. • Deliver the capstone: meet every ENSE 400 and ENSE 477 deliverable on schedule.
Project Objectives	Each objective is specific, measurable and dated. The KPI shows how it will be measured. 1. Page-anchored explanations. By 17 November 2026, a student can open a PDF from a local course folder, select a passage and receive an explanation with its page reference. • KPI: 100% of explanations show a page reference; median response time of 10 seconds or less on the free model, across a 20-question test set. 2. Accurate answers. By 24 November 2026, explanations are accurate to the source page. • KPI: at least 85% of 30 sampled explanations rated accurate by the team and two classmates, checked against the page. 3. Practice from the passage. By 1 December 2026, the app generates at least 5 flashcards and a 5-question quiz from a selected passage. • KPI: generation takes 20 seconds or less; at least 80% of generated cards rated correct and useful by testers. 4. Usable by students. Test the prototype with at least 5 students on 20 October 2026, and the working app with at least 5 students by 1 December 2026. • KPI: System Usability Scale (SUS) score of 70 or higher; at least 80% of test tasks completed without help. 5. Less setup time. By 1 December 2026, reduce the time from opening a file to receiving a first useful answer compared with the student's current method (uploading to a general AI assistant). • KPI: at least 50% less time, measured in usability sessions with at least 5 students. 6. Low running cost. Throughout both terms, keep AI spending within the self-funded budget. • KPI: US\$0 spent on free-tier requests; paid-model testing at or below US\$50.

	7. Early adoption and willingness to pay. By the end of ENSE 477 (April 2027), run a pilot with at least 15 design-partner students for at least 3 weeks. • KPI: at least 60% of pilot students active each week; at least 20% say they would pay for Pro.
Project Budget	Self-funded by the team (bootstrapped). No outside funding. • Paid AI models for testing: up to US\$50 across both terms. Free-tier AI models cost US\$0. • Hosting, accounts and database: US\$0 on free tiers. • Design and code tools: US\$0 (Figma education plan, GitHub, VS Code). • Apple Developer Program: US\$99 a year, only if the iPad version goes ahead in ENSE 477. Total: up to US\$150 over two terms.
Project Sponsor	Aubin Chriss Izere and Chop Kur, co-founders. The project is self-funded by the team.
Project Manager	Aubin Chriss Izere and Chop Kur, co-project managers. Project management is shared and the work is split by feature: each member plans, builds and tests the features they own, and both review the schedule and risks together each week.
Additional Key Project Stakeholders	
• Dr. Tim Maciag, ENSE 400/477 instructor: reviews and grades deliverables and runs check-ins and scrums. • Linda, U of R student: interview participant whose problem statement shaped the project. • Design-partner students, U of R classmates: test the app on their own course files and give regular feedback. • ENSE 400 capstone peers: usability testers at the 20 October session and audience at pitches and bazaar day. • Cultivator HQ mentors: audience for the 13 October final pitch; startup and business feedback. • U of R undergraduate students: primary end users. • Graduate and thesis students: secondary end users. • AI model providers (for example Google, Groq, OpenRouter and Anthropic): supply the free and paid models; their pricing, rate limits and terms affect the product. • Course instructors and textbook publishers: own the course materials students study from.	
Overall Project Milestones	Dates
Project idea pitched and selected (complete)	15 Sep 2026

Planning documents complete (Workshop 3); project review and GitHub check-in	6 Oct 2026
Final pitch at Cultivator HQ; Vlog 1 submitted	13 Oct 2026
Low- and high-fidelity prototypes and usability questionnaire ready; usability evaluation session	20 Oct 2026
Usability results analysed; redesign and system architecture done; development starts	27 Oct 2026
Local course folder access and PDF reader with passage selection working	3 Nov 2026
AI explanations with page references and the model picker working	17 Nov 2026
Work-in-progress demo at bazaar day	24 Nov 2026
Flashcards and quizzes working; ENSE 400 scope complete; Vlog 2 submitted	1 Dec 2026
User accounts, Pro subscription and payments (tentative)	Late Jan 2027
Spaced review against exam dates and voice input (tentative)	Late Feb 2027
iPad version beta; design-partner pilot starts (tentative)	Mid Mar 2027
Final release and capstone showcase (tentative)	Early Apr 2027

Overall Project Risks

- **Free AI tiers change:** providers can lower rate limits, change terms or end free access. Free-tier limits are also shared by every student using the team's key.
- **Privacy:** selected passages are sent to outside AI providers, and some free tiers allow prompts to be used to improve their models. This must be disclosed clearly to students.
- **Wrong answers:** AI explanations or page references may be inaccurate.
- **Unreadable PDFs:** scanned or image-only PDFs have no text layer, so passages cannot be selected or sent without text recognition (OCR).
- **Subscription cost:** heavy use of the most expensive models could cost more than a Pro subscription brings in without monthly usage limits.

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- **Scope:** accounts, payments, voice input and an iPad version are a lot for a two-person team over two terms.
 - **Team availability:** with only two members, illness or other course deadlines can stall a whole feature.
 - **Adoption:** students may stay with free tools they already use, such as ChatGPT or NotebookLM.
 - **Copyright:** course materials belong to instructors and publishers; the app must send only the selected passage and never store or share course files.
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